



MEASURING INSTRUMENTS & SCALES — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. SONAR & Sound Instruments — Underwater Detection and Sound Intensity
2. Wind & Survey Instruments — Anemometer and Tacheometer
3. Electrical Measuring Instruments — Ammeter, Voltmeter, Potentiometer & Galvanometer
4. High Temperature & Solar Radiation — Pyrometer and Pyrheliometer
5. Pressure & Atmospheric Instruments — Manometer, Barometer & Barograph
6. Vehicle Distance & Precision Time — Odometer, Speedometer & Chronometer
7. Density of Milk & Liquids — Lactometer, Butyrometer & Hydrometer
8. Humidity & Atmospheric Moisture — Hygrometer and Related Devices
9. Special Scales & Temperature Sensors — Ringelmann Scale and Thermoresistor
10. Medical, Light & Purity Instruments — EEG, Sphygmomanometer, Luxmeter & Polygraph
11. RADAR — Radio Detection and Ranging
12. Earthquake Measurement — Richter, Mercalli, Seismograph & Seismometer
13. Depth, Altitude & Surveying — Fathometer, Altimeter & Theodolite
14. Quick Match-the-Following Instrument Map
15. Phonometer — Disputed Source Question
16. Accelerometer — Crash, Free-Fall and Smartphone Tilt Detection

1. SONAR & Sound Instruments — Underwater Detection and Sound Intensity

- SONAR ka full form Sound Navigation and Ranging hai. Ocean mein submerged objects, submarines aur icebergs locate karne ke liye iska use hota hai. [Q1]
- SONAR ka practical use mainly navigators/naval navigation mein hota hai. [Q2]
- SONAR ultrasonic waves use karta hai: pulse water mein bheji jati hai, target se reflect hokar echo return karta hai; time delay se location/depth estimate hoti hai. [Q3]
- Audiometer sound ki intensity/hearing level measure karta hai. Audiophone ek hearing-aid type device hai; chronometer highly accurate timepiece hota hai. [Q4]

2. Wind & Survey Instruments — Anemometer and Tacheometer

- Anemometer wind ki speed/velocity aur wind pressure/force ko measure karta hai. [Q5, Q6, Q7]
- Correct instrument map: Anemometer → wind speed; Ammeter → electric current; Pyrometer → high temperature. [Q8]
- Tacheometer ek rapid-surveying instrument/theodolite type device hai jo distance aur related survey measurements ke liye use hota hai; pressure difference measure nahi karta. [Q8]

EXAM TRAP — Tacheometer ≠ Pressure Difference — Q8

- ▶ Tacheometer rapid surveying/distance measurement ke liye hota hai; pressure difference ke saath pair incorrect hai. [Q8]
- ▶ Pressure difference ke liye manometer/differential manometer type instruments use kiye jate hain. [Q8]

3. Electrical Measuring Instruments — Ammeter, Voltmeter, Potentiometer & Galvanometer

- Ammeter circuit mein electric current measure karta hai; current ki SI unit ampere (A) hai. [Q9]
- Voltmeter → potential difference; Potentiometer → EMF/potential comparison; Galvanometer → very small electric current/detection. [Q10]
- Galvanometer electric resistance measure karne ka standard instrument nahi hai; resistance ke liye ohmmeter use hota hai. [Q10]
- Ammeter electric power measure nahi karta. Electric power wattmeter se measure ki jati hai. [Q19]

EXAM TRAP — Ammeter, Galvanometer & Power — Q9–Q10, Q19

- ▶ Ammeter → current; Galvanometer → small current; resistance → ohmmeter. Inhe interchange mat karein. [Q9, Q10]
- ▶ Electric power → wattmeter, not ammeter. [Q19]

4. High Temperature & Solar Radiation — Pyrometer and Pyrhelimeter

- Pyrometer very high temperature measure karne wala radiation/remot-sensing thermometer hai; furnaces ya inaccessible hot surfaces ke liye useful hota hai. [Q11, Q12, Q13, Q15]
- Pyrometer ka principle thermal radiation se linked hai; black-body radiation ke context mein Stefan–Boltzmann relation $E \propto T^4$ relevant hai. [Q11]
- Around 2000°C jaise very high temperature ke liye total radiation pyrometer suitable hota hai. [Q14]
- Pyrhelimeter direct solar radiation measure karta hai; pyranometer bhi solar irradiance measurement mein use hota hai. [Q16]

EXAM TRAP — Pyrometer Range — Q11–Q15

- ▶ Pyrometer high/very-high temperature ke liye hai; source repeatedly 1500°C se above temperature ke liye Pyrometer answer deta hai. [Q11, Q12, Q15]
- ▶ 2000°C context mein Total Radiation Pyrometer choose karna hai. [Q14]

5. Pressure & Atmospheric Instruments — Manometer, Barometer & Barograph

- Manometer liquid/gas pressure measure karta hai, commonly liquid-column pressure measurement principle par. [Q17, Q49]
- Barometer atmospheric pressure measure karta hai. Simple barometer traditionally mercury column use karta hai. [Q21, Q22]
- Barograph atmospheric pressure ko continuously record karta hai. [Q48]
- Useful matching: Ammeter → electric current; Hygrometer → relative humidity; Spring balance → weight; Barometer → atmospheric pressure. [Q18]

6. Vehicle Distance & Precision Time — Odometer, Speedometer & Chronometer

- Odometer vehicle dwara travelled distance measure karta hai; Speedometer vehicle ki instantaneous speed batata hai. [Q20]
- Chronometer exceptionally accurate timepiece hai, historically maritime navigation mein longitude determination ke liye important raha hai. [Q20]

7. Density of Milk & Liquids — Lactometer, Butyrometer & Hydrometer

- Lactometer milk ki density/specific gravity measure karta hai. [Q23, Q24]
- Butyrometer milk/milk products ka fat content measure karta hai, jabki Hydrometer liquids ki relative density/specific gravity measure karta hai. [Q23, Q25]

8. Humidity & Atmospheric Moisture — Hygrometer and Related Devices

- Hygrometer atmosphere/air ki humidity ya relative humidity measure karta hai. [Q25, Q26, Q27, Q28, Q29]
- Spectrometer electromagnetic spectrum/light properties analyze karta hai; Eudiometer gas-mixture volume change measure karta hai. [Q26]
- Hypsometer height/altitude estimation ke liye use ho sakta hai; Thermostat kisi system ka temperature desired set-point ke around regulate karta hai. [Q29]

9. Special Scales & Temperature Sensors — Ringelmann Scale and Thermoresistor

- Ringelmann Scale smoke ki density/opacity ko grade karne ke liye use hota hai; smoke-abatement work mein iska classic use hai. [Q30]
- Thermoresistor/thermistor electronic temperature sensing mein use hota hai; temperature change ke saath electrical resistance change hoti hai. [Q31]

10. Medical, Light & Purity Instruments — EEG, Sphygmomanometer, Luxmeter & Polygraph

- EEG (Electroencephalogram) scalp electrodes ke through brain ki electrical activity monitor/record karta hai. [Q32]
- Stethoscope → heart/body sounds; Sphygmomanometer → blood pressure; Caratometer → gold purity; Luxmeter → light intensity/illuminance. [Q33, Q34, Q35]
- Polygraph ya lie detector blood pressure, pulse, respiration aur skin conductivity jaise physiological indices record karta hai. [Q51]
- Kymograph pressure/tension variations record kar sakta hai; Gyroscope orientation/rotation sensing device hai; Pyrometer high surface temperature measure karta hai. [Q51]

11. RADAR — Radio Detection and Ranging

- RADAR ka full form Radio Detection and Ranging hai aur ye radio waves par based hota hai. [Q36]
- RADAR objects ki presence/location ke saath range, angle aur velocity determine kar sakta hai; aircraft, ships, missiles, weather formations etc. detect kiye ja sakte hain. [Q36]

12. Earthquake Measurement — Richter, Mercalli, Seismograph & Seismometer

- Exam-source convention mein Richter scale ko earthquake ki intensity/magnitude se associate kiya gaya hai; scale seismic-wave amplitude aur released-energy related magnitude ko represent karta hai. [Q37, Q38, Q39]
- Richter scale humidity measure nahi karta; humidity ke liye Hygrometer hota hai. [Q40]
- Seismograph earthquake-generated seismic waves ko detect/record karta hai; source questions mein earthquake measurement/recording ke liye isi ko answer diya gaya hai. [Q41, Q42, Q43, Q44]
- Seismometer seismic waves measure karta hai; “surface curvature” ke saath iska matching incorrect hai. [Q49]

⚠ EXAM TRAP — Richter Magnitude vs Mercalli Intensity — Q37–Q40

- ▶ Scientific distinction: Richter-type scales earthquake MAGNITUDE/energy-seismic amplitude se related hain; Mercalli scale observed INTENSITY/damage effects describe karta hai. [Q37, Q39]
- ▶ Lekin source/commission ke kuch questions “intensity” wording ke saath Richter ko expected answer dete hain; exam mein source convention ko recognize karein. [Q37, Q38]

13. Depth, Altitude & Surveying — Fathometer, Altimeter & Theodolite

- Altimeter kisi fixed reference level ke above altitude/height measure karta hai. [Q45, Q46]
- Fathometer ultrasonic waves ke through sea/water depth ya submerged object ki depth determine karta hai. [Q46, Q47]
- Theodolite horizontal plane (azimuth) aur vertical plane (elevation) mein angular distances/angles measure karta hai. [Q52]

14. Quick Match-the-Following Instrument Map

- Core matching: Earthquake → Seismograph; Height → Altimeter; Electric current → Ammeter; Electrical resistance ki unit → Ohm. [Q45]

- Core matching: Fathometer → depth of sea; Barometer → atmospheric pressure; Hygrometer → atmospheric humidity; Altimeter → height. [Q46]
- Core matching: Anemometer → wind velocity; Seismograph → earthquake; Barograph → atmospheric pressure; Hygrometer → humidity. [Q48]
- Other correct pairs: Manometer → pressure; Carburetor → air-fuel mixing in IC engine; Cardiograph → heart movement/recording. [Q49]

15. Phonometer — Disputed Source Question

- Phonometer human voice ki force/strength test karne se associated instrument hai; Photometer light intensity measure karta hai. [Q50]
- Source question ke diye options mein koi option fully correct nahi tha; official answer “* / no correct option” marked hai. [Q50]

EXAM TRAP — Phonometer Question Is Disputed — Q50

- ▶ Phonometer → force/strength of human voice; Photometer → light intensity. Dono alag instruments hain. [Q50]
- ▶ Given options mein correct phonometer function absent tha, isliye official answer “*/no correct option” hai. [Q50]

16. Accelerometer — Crash, Free-Fall and Smartphone Tilt Detection

- Accelerometer acceleration/vibration of motion measure karta hai; MEMS accelerometers modern electronic devices mein common hain. [Q53]
- Car crash mein rapid deceleration detect karke airbag deployment trigger karna accelerometer application hai. [Q53]
- Laptop free-fall detection aur smartphone portrait/landscape tilt sensing bhi accelerometer se hota hai; source ke tino actions correct hain. [Q53]